# SLIC Segmentation Algorithm: Performance Optimization through Concurrent Programming with OpenMP

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# Introduction to SLIC Segmentation



- **Simple Linear Iterative Clustering (SLIC) Segmentation:** Computer Vision Algorithm.
- **Goal:** Groups pixels into superpixels similar in color and proximity, preserving boundaries.
- **Problem:** Sequential execution is a bottleneck for high-resolution images.
- **Complexity:** $O(N)$.
- **Challenge:** Accelerate the algorithm using CPU Multithreading with OpenMP.

# SLIC Segmentation Example

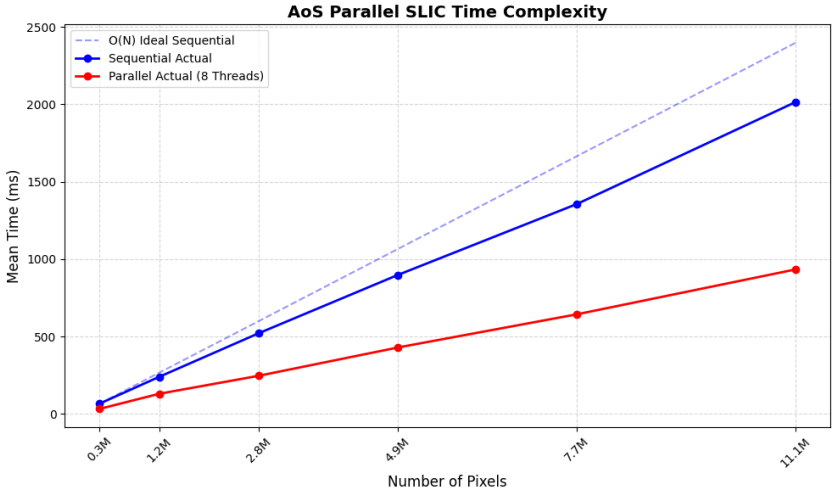
## 1 Introduction

Figure: Original Image

Figure: Segmented Image

# Complexity of the Algorithm





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2 Approach

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# Experiment Setup & Baseline

## 2 Approach

- **Hardware Environment:**
  - **CPU:** Apple Silicon M1 ARM (8 Cores).
  - **Topology:** Asymmetric (4 P-Cores + 4 E-Cores).
  - **Cache:** 128-byte Cache Line size on P-Cores.
  - 8 GB Unified Memory
- **Software & Tools:**
  - **Language:** C++ & OpenMP.
  - **Compiler Flags:** `-O3`, `-ffast-math`.
  - **Analysis & Plotting:** Python, Matplotlib.
- **Dataset:**
  - Berkeley Segmentation Dataset ($640 \times 480$).
  - Target Superpixels ($K$): 1000.

# Target Phases to Parallelize

## 2 Approach

- **Initialization (Seq):** Grid setup and gradient perturbation.
- **Assignment (Parallel):** Find the nearest cluster center for each pixel.
- **Update Centroids (Parallel):**
  - Recalculate cluster centers.
  - High risk of *Race Conditions*.
- **Enforce Connectivity (Seq):**
  - Merges disjoint regions (BFS approach).
  - Kept sequential to avoid massive synchronization overhead.

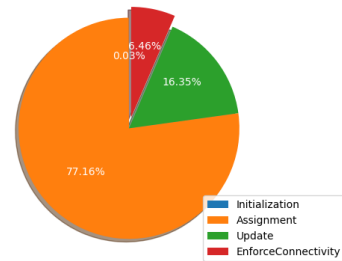


Figure: Parallelizable vs Sequential Workload

# Architectural Tuning & Optimization

## 2 Approach

- **NUMA-Aware Initialization (First-Touch):**
  - Parallel array initialization physically maps memory to the local core cluster, reducing latency.

- **Data Layout & Vectorization:**
  - **SoA:** Enables `-O3` auto-vectorization for color computations.
  - **AoS:** Explicit `alignas(128)` padding eliminates False Sharing.
  - **Explicit SIMD:** `#pragma omp simd` accelerates centroid averaging.

- **2D Cache Tiling ($2S \times 2S$):**
  - Pins local centroids in L1/L2 caches to maximize **Temporal Locality**.

# Assignment Method

## 2 Approach

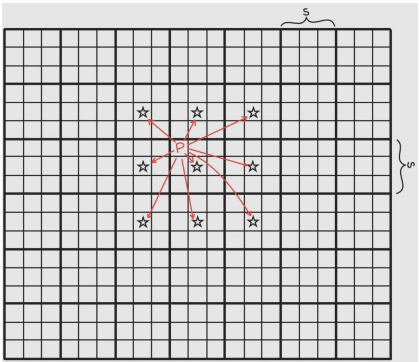


Figure: Pixel Centric (Chosen)

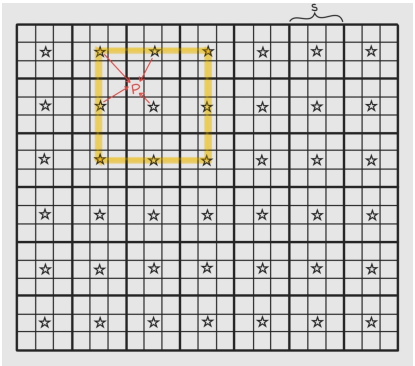


Figure: Centroid Centric (Discarded)

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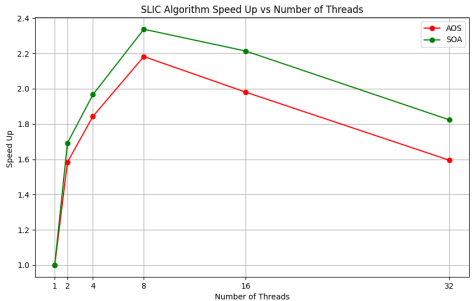
# Thread Scalability & Oversubscription

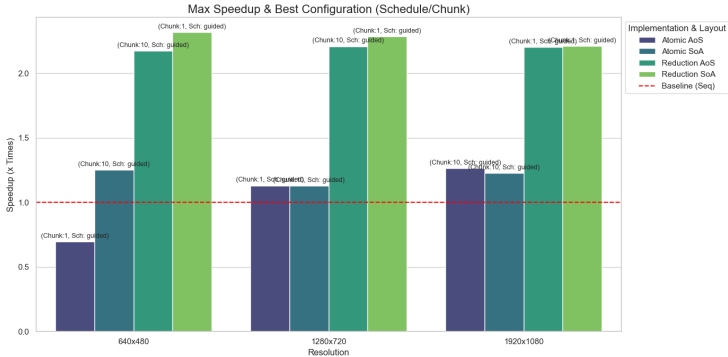
## 3 Experiments




- **The "Sweet Spot" (8 Threads):**
  - Peak strictly matches the hardware physical cores (no Hyper-Threading).
- **Hardware Oversubscription ($> 8$):**
  - **Context Switching:** OS wastes cycles swapping thread states.
  - **Cache Thrashing:** Suspended threads lose their L1 data.
  - **AoS Padding Penalty:** The `alignas` padding in AoS amplifies the memory bandwidth waste during thrashing.

# Best Synchronization & Scheduling

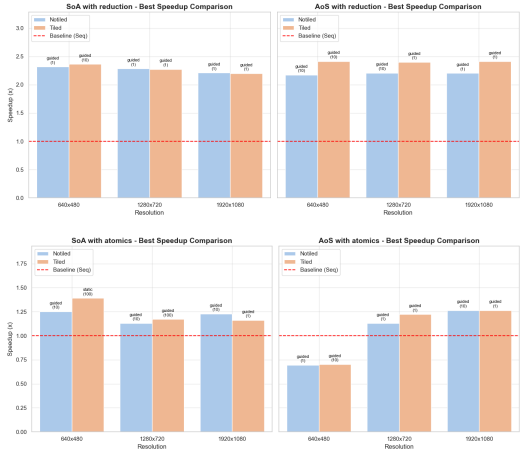
## 3 Experiments




- **Reductions** > **Atomics:** Atomics suffer from *True Sharing* on centroid accumulators, stalling the CPU pipeline. Reductions use thread-private workspaces.
- **Guided Scheduler:** Perfectly balances the asymmetric workload between P-Cores and E-Cores.

# The Tiling Showdown (2D Cache Blocking)

## 3 Experiments




- **Goal:** Use $2S \times 2S$ Tiling in the Assignment Phase to pin the 9 local centroids in the L1/L2 cache.
- **AoS Resilience (The Winner):**
  - *Perfect Marriage:* Centroids stay pinned, while "tourist" pixels are loaded contiguously ($L, a, b, x, y$ in one fetch).
- **SoA Collapse:**
  - *Structural Fragmentation:* Scattered across separate arrays.
  - 2D strided access breaks the hardware prefetcher.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4 Conclusions

# Final Conclusions

## 4 Conclusions

- **The Winning Configuration (2.41x Speedup):**
  - **AoS Layout + Tiling + Reductions + Guided Scheduler.**
  - Maximizes spatial/temporal locality and handles the P/E Core asymmetry without True Sharing.
- **Memory Access Dictates Layout:**
  - **SoA** is strictly superior for *Linear Scans* (relies on HW Prefetcher).
  - **AoS** is strictly superior for *2D/Strided Access* (relies on pixel-level spatial locality).
- **Amdahl's Limit:**
  - The 6% sequential time (Initialization & Enforce Connectivity) caps the maximum theoretical speedup at $\sim 5.6\times$ on 8 cores.
  - Achieving $2.41\times$ indicates the bottleneck is hardware memory bandwidth, not the sequential phases.
- **Future Works:** GPU Porting via CUDA (massive memory bandwidth) and Centroid-Centric approach exploration.

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# Thank you for your attention!

Any Questions?